

Indicators list

*A lot of this variables are extracted thanks to: [https://scikit-](https://scikit-image.org/docs/0.24.x/auto_examples/segmentation/plot_regionprops.html)

[image.org/docs/0.24.x/auto_examples/segmentation/plot_regionprops.html](https://scikit-image.org/docs/0.24.x/auto_examples/segmentation/plot_regionprops.html)

Variables	Calculation
Number	Index of the school among all the images
Label	Index of the school among one image (1 if only 1 school in the image)
File_name	Name of the resize matrix used for the school extraction
Size	Number of pixels of the school
Bbox	Box around the school*
Center	Centroid*
Depth	Depth of the average coordinate of the school in meter
Sea_depth	Sea floor depth in meter
School_seabed_distance	Depth- Sea_Depth in meter
Gps_lon_lat	Longitude and latitude (WGS84)
Time	Time UTC
Width_bbox	Width Bbox*
Length_box	Length Bbox*
Is_very_wide	<code>(bbox[3]-bbox[1])>width_image/2</code>
Is_very_tall	<code>(bbox[2]-bbox[0])>height_image/2</code>
Dis_to_surface	Bbox[0]* in pixels
Dis_level	0,1 or 2 0 if Dis_to_surface under 10 pixels, 1 if between 10 and 50 and 2 if > 50
Axis_major_length	The length of the major axis of the ellipse that has the same normalized second central moments as the region*
Axis_minor_length	The length of the minor axis of the ellipse that has the same normalized second central moments as the region.*
Perimeter_school	Perimeter*
Intensity_school	Mean intensity of the school
Intensity_img	Mean intensity of the image
Dif_intensity_school_image	Intensity_school – Intensity_img
Center_square_intensity	Center square: A Square of nine pixels with central pixel is the centroid* of the school Mean intensity among this square
Edges_school_intensity	Edges: List of tuples containing coordinates of the four connected neighbours Mean intensity among these coordinates
Dif_intensity_center_edges	Cente_square_intensity - Edges_school_intensity
Std_intensity_school	Intensity standard deviation of the school
Gradient_school	<code>gradient_y, gradient_x = np.gradient(coords) #coords of the school</code> <code>gradient_magnitude = np.sqrt(gradient_x**2 + gradient_y**2)</code> <code>gradient_school = np.mean(gradient_magnitude)</code>

Gradient_school_center	Idem than before but for the coord of the central square
Gradient_school_edges	Idem than before but for the coord of the edges
Dif_gradient_center_edges	Gradient_school_center -Gradient_school_edges
Width_length_ratio	<code>region.bbox[3]-region.bbox[1])/(region.bbox[2]-region.bbox[0]</code>
Axis_ellipse_ratio	<code>region.axis_minor_length / region.axis_major_length</code>
Solidity	Ratio of pixels in the region to pixels of the convex hull image.*
Compactness	<code>4*math.pi*(region.area)/(region.perimeter**2)</code>
Inertia_tensor_eigvals_ratio	<code>region.inertia_tensor_eigvals[1]/region.inertia_tensor_eigvals[0]</code>
Perimeter_area_ratio	<code>region.perimeter/region.area</code>
Thresh_min	Thresh_min from double thresholding
Thresh_max	Thresh_max from double thresholding
Dis_to_bottom	Same than School_seabed_distance (But I just found an error in the calculation you can remove it and only keep School_seabed_distance)